



Improving Customer Data Accuracy in the Retail Industry with ML-Based Deduplication

Case Study



Estimated Annual Business Impact

\$1.2M+

In annual savings from reduced duplicate outreach, analyst effort, and campaign waste.

30-40%

Increase in campaign conversions through improved personalization and targeting.

70%

Improve ROI from Ad spent and saved analyst time in CRM management





Business Problem

Retailers rely heavily on accurate customer data to drive personalization, loyalty programs, and omnichannel experiences. However, poor data quality has a significant operational and financial impact:



Over 10% of customer records in CRM systems are duplicates, causing fragmented purchase histories.



Traditional rule-based deduplication misses 30–40% of potential matches due to typos, nicknames, and inconsistent formatting.



Marketing teams waste 20–30% of campaign budgets on duplicate outreach and redundant mailers.



Manual cleanup consumes 100+ analyst hours per month, delaying campaign launches and analytics.



Data fragmentation leads to poor personalization, resulting in 15–25% lower conversion rates.

High-Level Solution

ML-BASED CUSTOMER MATCHING

- Applied fuzzy string similarity (e.g., Jaro-Winkler, Levenshtein) across name, address, email, and phone fields.
- Used ML classification models trained on labeled examples to identify duplicates dynamically.

HUMAN-IN-THE-LOOP ACTIVE LEARNING

- Escalated only uncertain pairs to analysts, reducing manual review by over 60%.
- Model improved iteratively as new labels were added.

GOLDEN PROFILE CREATION

Merged duplicates into single 'Golden Profiles' synchronized across CRM, loyalty, and analytics systems.

TECHNOLOGY STACK

- Python + Dedupe: Deduplication engine.
- Snowflake/BigQuery: Centralized data platform.
- Airflow: Automation of deduplication workflows.
- Streamlit: Analyst validation UI.
- Salesforce, Adobe, Klaviyo: Marketing and personalization platforms.

Retail

Impact Metrics

Impact Area	Before	After	Business Impact
Duplicate Customer Profiles	8–12% of CRM	<1.5%	↑ Accurate 360° view of customers
Manual CRM Cleanup Time	100+ hrs/month	<30 hrs/month	↓ 70% analyst time saved
Campaign Overlap Rate	~18%	<5%	↓ Wasted ad spend by 70%
Customer Lifetime Value (CLV) Accuracy	Low	High	↑ Precision in loyalty segmentation
Personalization Accuracy (emails, offers)	~60%	>90%	↑ CTR, conversion rates
Return-to-Campaign Attribution	Inconsistent	Reliable	↑ Clarity in ROI measurement



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